

Unit 13

Integration

Basic

1. Integrate $\frac{3}{x^2} + \frac{4}{3x-5}$ with respect to x . [4]
[N21/I/4]

Intermediate

2. A curve is such that $\frac{d^2y}{dx^2} = 2e^{-x} + 6e^{2x}$. The curve intersects the y -axis at $P(0, 5)$ and has a gradient of 3 at P . Find the equation of the curve. [7]
[N22/I/6]
3. (i) Differentiate $x^2 \ln x$ with respect to x . [2]
(ii) Hence find $\int x \ln x \, dx$. [3]
[N19/II/1]
4. $f(x)$ is such that $f''(x) = 3 \cos 3x - 4 \sin 2x$. Given that $f(0) = 0$ and $f\left(\frac{\pi}{2}\right) = \frac{5}{6}$, show that $f\left(\frac{\pi}{3}\right) = 1 + \frac{\sqrt{3}}{2}$. [9]
[N19/II/5]
5. It is given that $f(x)$ is defined for $x > \frac{3}{2}$ and is such that $f'(x) = \frac{10x-9}{2x-3}$.
(i) Express $f'(x)$ in the form $a + \frac{b}{2x-3}$, where a and b are constants. [2]
(ii) Hence explain whether $f(x)$ is an increasing or decreasing function. [1]
(iii) By considering $f''(x)$, explain whether $f'(x)$ is an increasing or decreasing function. [4]
(iv) Given that $f(2) = 8$, obtain an expression for $f(x)$. [4]
[N18/I/11]
6. (i) Differentiate $x \cos x$ with respect to x . [2]
(ii) Hence find $\int x \sin x \, dx$. [2]
(iii) Differentiate $x^2 \sin x$ with respect to x and hence, using the result found in part (ii), find $\int x^2 \cos x \, dx$. [6]
[N18/II/6]
7. A curve is such that $\frac{d^2y}{dx^2} = 8 - 6x$ and the point $P(2, 8)$ lies on the curve. The gradient of the curve at P is 3. Find the equation of the curve. [6]
[N17/I/1]
8. (i) Show that $\frac{d}{dx} \left(\frac{\ln x}{x^2} \right) = \frac{1}{x^3} - \frac{2 \ln x}{x^3}$. [3]
(ii) Integrate $\frac{\ln x}{x^3}$ with respect to x . [3]
(iii) Given that the curve $y = f(x)$ passes through the point $\left(1, \frac{3}{4}\right)$ and is such that $f'(x) = \frac{\ln x}{x^3}$, find $f(x)$. [2]
[N16/II/10]

9. The curve $y = f(x)$ is such that $f'(x) = 2e^x + e^{-2x}$.

(i) Explain why the curve $y = f(x)$ has no stationary points. [2]

(ii) Given that the curve passes through the point $(0, 2)$, find an expression for $f(x)$. [4]

[N15/II/1]

Advanced

10. The function f is defined for $x \in \mathbb{R}$ and is such that $f''(x) = 48x^2 + 2e^{2x-1}$.

The line $x = \frac{1}{2}$ is the normal to the curve $y = f(x)$ at the point where $y = \frac{1}{4}$.

(i) Find an expression for $f'(x)$. [5]

(ii) Hence find an expression for $f(x)$. [4]

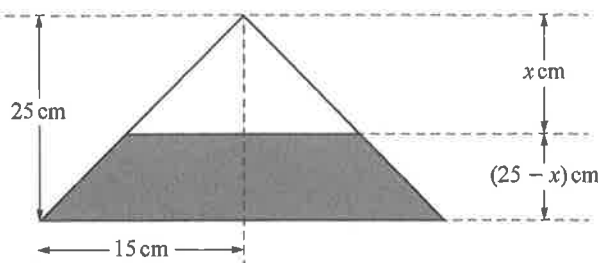
(iii) Show that the equation of the tangent to the curve $y = f(x)$ at the point where the curve intersects the y -axis can be written as

$$2e(y + 3x - 1) = 2x + 1.$$

[3]

[N20/II/10]

11. [The volume of a cone of height h and base radius r is $\frac{1}{3}\pi r^2 h$.]



The diagram shows a vertical cross-section of a container in the form of a cone, height 25 cm and base radius 15 cm. The container is initially empty. At time $t = 0$ liquid is allowed to flow in through a small hole close to the vertex. After t seconds the height of liquid in the container is $(25 - x)$ cm and the volume of liquid is V cm³.

(i) Show that $V = 3\pi \left(625 - \frac{x^3}{25} \right)$. [3]

Given that the rate of flow of the liquid is kx^2 cm³/s, where k is a constant,

(ii) find an expression for $\frac{dt}{dx}$ in terms of π and k . [4]

The liquid was allowed to flow for 72π seconds when the height of the liquid reached 12 cm.

(iii) By expressing t as a function of x , find the value of k . [4]

[N20/II/11]